

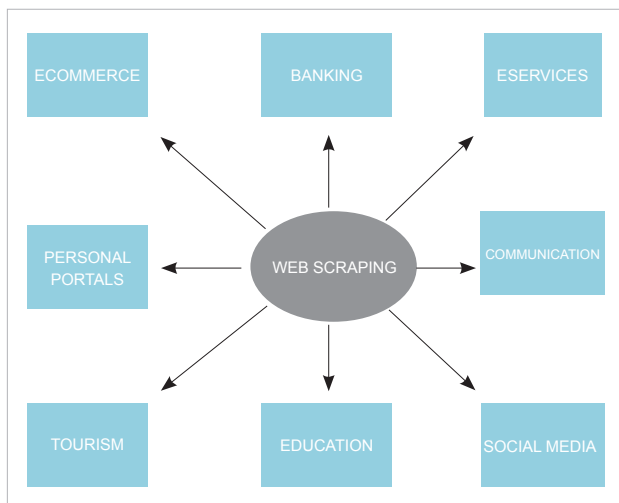


Figure 1: Web scraping as a service

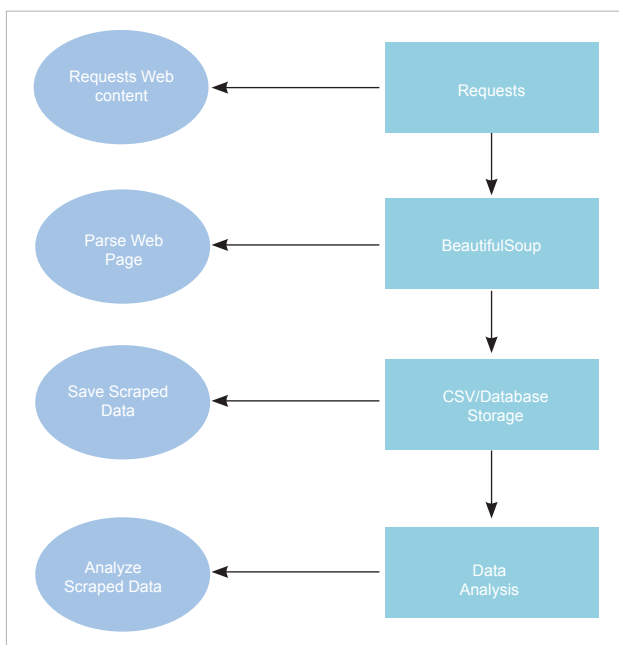


Figure 2: Schematic diagram of different stages of web scripting

Step 3: Storing the data

Subsequently, the web scraper stores the extracted data in CSV, JSON, or a database.

Step 4: Analysing the data

From these well-established formats, the web scraper initiates analysis of the stored data.

Web scraping of Python can be initiated with two simple library modules — *requests* and *BeautifulSoup*. In this article, we will demonstrate their functions as well as Python web scripting, with a few illustrations to help you understand the basics of web scraping.

requests module

Making HTTP requests and receiving responses are done with the help of the *requests* library. The management of the request and response is handled by built-in functionality in Python requests. Since this module is not shipped with core Python it is necessary to install it with *pip* separately.

`pip install requests`

Making a GET request

The Python *requests* module comes with some built-in ways to send GET, POST, PUT, PATCH, or HEAD requests to a given URI. An HTTP request may send data to a server or get it from a specified URI. It functions as a client-server request-response protocol. Information is retrieved from the specified server using the GET method and the given URI. The user data is encoded and added to the page request before being sent via the GET method.

```

import requests
# Making a GET request in Python
r = requests.get('https://www.tutorialspoint.com/python-for-beginners-python-3/index.asp')
# check the status code for the response received
# success code - 200
print(r.status_code)
#print header
print(r.headers)
  
```

The output is:

```

200
{'Content-Encoding': 'gzip', 'Accept-Ranges': 'bytes',
'Access-Control-Allow-Credentials': 'true', 'Access-Control-
Allow-Headers': 'X-Requested
-With, Content-Type, Origin, Cache-Control, Pragma,
Authorization, Accept, Accept-Encoding', 'Access-Control-
Allow-Methods': 'GET, POST, PUT, P
ATCH, DELETE, OPTIONS', 'Access-Control-Allow-Origin': '*',
*;', 'Access-Control-Max-Age': '1000', 'Age': '515359',
'Cache-Control': 'max-age=25
92000', 'Content-Type': 'text/html; charset=UTF-8', 'Date':
'Thu, 25 May 2023 08:05:00 GMT', 'Expires': 'Sat, 24 Jun 2023
08:05:00 GMT',
Modified': 'Fri, 19 May 2023 08:55:43 GMT', 'Server': 'ECCac
(blr/D141)', 'Vary': 'Accept-Encoding', 'X-Cache': 'HIT',
'X-Frame-Options': 'SAME
ORIGIN, SAMEORIGIN', 'X-Version': 'OCT-10 V1', 'X-XSS-
Protection': '1; mode=block', 'Content-Length': '14119'}
  
```

The response 'r' is a powerful object with a plethora of methods and features that aid in normalising data or offer the